

For example:

$$2x - 2 \text{ (Substitute when } x = 2)$$

$$\rightarrow 2(2) - 2$$

$$= 4 - 2 = 2$$

2 Algebra and graphs

C2.1 Introduction to algebra

Notes and examples

- 1 Know that letters can be used to represent generalised numbers.
- 2 Substitute numbers into expressions and formulas.

→ Variables such as: x, y, w, z , etc.
Most commonly used variables: x and y

Difference between expressions and equations:

Expression: Does not contain an equal sign.
E.g. $2x - 2$

Equation: Contains an equal sign.
E.g. $2x - 2 = 0$

C2.2 Algebraic manipulation

Notes and examples

- 1 Simplify expressions by collecting like terms.

Simplify means give the answer in its simplest form,

$$\text{e.g. } 2a + 3b + 5a - 9b = 7a - 6b.$$

- 2 Expand products of algebraic expressions.

E.g. $(2x - 1)(x + 2)$

$$2x^2 + 4x - x - 2 = 2x^2 + 3x - 2$$

$$(2x)(x) + (2x)(2) + (-1)(x) + (-1)(2)$$

- 3 Factorise by extracting common factors.

E.g. $2x^2 + 3x - 2$

$$= 2x^2 + 4x - x - 2$$

$$= 2x(x + 2) - 1(x + 2)$$

$$= (x + 2)(2x - 1)$$

Bigger value gets the sign of the original value and the other value is adjusted accordingly.

C2.3 Extended content only.

C2.4 Indices II

Notes and examples

- 1 Understand and use indices (positive, zero and negative).

- 2 Understand and use the rules of indices:

$$\begin{aligned} ① a^x \times a^y &= a^{x+y} & ⑥ a^{-x} &= \frac{1}{a^x} \\ ② a^x \div a^y &= a^{x-y} & ⑦ a^{\frac{x}{y}} &= \sqrt[y]{a^x} \\ ③ (a^x)^y &= a^{xy} \\ ④ a^0 &= 1 \\ ⑤ a^1 &= a \end{aligned}$$

e.g. $2^x = 32$. Find the value of x .

$$2 \times 2 \times 2 \times 2 \times 2 = 32 = 2^5$$

e.g. simplify: $\frac{16}{32} = \frac{2^4}{2^5} = \frac{1}{2} \therefore x = 5$

- $(5x^3)^2 = 5x^6$
- $12a^5 \div 3a^{-2} = 4a^7$
- $6x^7y^4 \times 5x^{-5}y = 30x^2y^5$

Knowledge of logarithms is **not** required.

Same signs = Subtract
Different signs = ADD

C2.5 Equations

Notes and examples

- 1 Construct simple expressions, equations and formulas.

And, formulas link one value to one or more other values.

- 2 Solve linear equations in one unknown.

- 3 Solve simultaneous linear equations in two unknowns.

- 4 Change the subject of simple formulas.

E.g. $y = 3x + 4$ (Make x the subject)

$$y - 4 = 3x + 4 - 4$$

$$\frac{y - 4}{3} = \frac{3x}{3}$$

$$\therefore x = \frac{y - 4}{3}$$

e.g. write an expression for a number that is 2 more than n . $n + 2$

Includes constructing linear simultaneous equations.

Examples include:

$$\begin{aligned} 3x + 4 &= 10 \rightarrow 3x + 4 - 4 = 10 - 4 \rightarrow \frac{3x}{3} = \frac{6}{3} \rightarrow x = 2 \\ 5 - 2x &= 3(x + 7) \rightarrow 5 - 2x - 3x - 21 = 3x + 21 - 3x - 21 \rightarrow 5 - 2x - 3x - 21 = 0 \end{aligned}$$

e.g. change the subject of formulas where:

- the subject only appears once
- there is **not** a power or root of the subject.

Difference between expressions and equations:

Expression: Does not contain an equal sign.
E.g. $2x - 2$

Equation: Contains an equal sign.
E.g. $2x - 2 = 0$

2 Algebra and graphs (continued)

C2.6 Inequalities

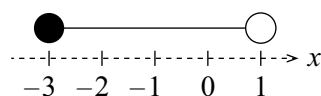
Represent and interpret inequalities, including on a number line.

Notes and examples

When representing and interpreting inequalities on a number line:

- open circles should be used to represent strict inequalities ($<$, $>$)
- closed circles should be used to represent inclusive inequalities ($\leq$, $\geq$)

e.g. $-3 \leq x < 1$



C2.7 Sequences

Notes and examples

1 Continue a given number sequence or pattern.

e.g. write the next two terms in this sequence:

1, 3, 6, 10, 15, 21, 28
 2 3 4 5 6
 It's advancing by $x+1$.

2 Recognise patterns in sequences, including the term-to-term rule, and relationships between different sequences. e.g. 1, 2, 3, 4, 5, ... (Term-to-term rule $\Rightarrow$ add 1)

3 Find and use the n th term of the following sequences:

(a) linear $n^{\text{th}} \text{ term} = a + (n-1)d$

(b) simple quadratic $n^{\text{th}} \text{ term} = an^2 + bn + c$ where e.g. find the n th term of 2, 5, 10, 17

(c) simple cubic $n^{\text{th}} \text{ term} = an^3 + bn^2 + cn + d$
 $2a = 2^{\text{nd}} \text{ difference}$
 $3a + b = 1^{\text{st}} \text{ difference (2nd term - 1st term)}$
 $6a = 3^{\text{rd}} \text{ difference}$
 $12a + 2b = 1^{\text{st}} \text{ term of 2nd difference}$
 $7a + 3b + c = 1^{\text{st}} \text{ difference (2nd term - 1st term)}$
 $a + b + c + d = 1^{\text{st}} \text{ term}$

C2.8 Extended content only.

$a + b + c + d = 1^{\text{st}} \text{ term}$

(d) Geometric sequence $n^{\text{th}} \text{ term} = ar^{n-1}$

Common ratio $\left(\frac{2^{\text{nd}} \text{ term}}{1^{\text{st}} \text{ term}} \right)$

Should be same for $\frac{3^{\text{rd}} \text{ term}}{2^{\text{nd}} \text{ term}}$ and so on as well.

C2.9 Graphs in practical situations

Notes and examples

1 Use and interpret graphs in practical situations including travel graphs and conversion graphs.

e.g. interpret the gradient of a straight-line graph as a rate of change. $\downarrow = \frac{y_2 - y_1}{x_2 - x_1}$ (between 2 points)

2 Draw graphs from given data.

e.g. draw a distance-time graph to represent a journey.

* Gradient of a speed-time graph = acceleration

* Gradient of a distance-time graph = speed

* Area under a speed-time graph = distance travelled

$$y = 2x + 3 = 2(-2) + 3$$

$$= -4 + 3 = -1$$

$$y = 2x + 3 = 2(-1) + 3$$

$$= -2 + 3 = 1$$

y	-2	-1	0	1	2
x	-1	1	3	5	7

Eg. $y = 2x + 3$

$$y = 2x + 3 = 2(0) + 3 = 3$$

$$y = 2x + 3 = 2(1) + 3$$

$$= 2 + 3$$

$$= 5$$

$$y = 2x + 3 = 2(2) + 3$$

$$= 4 + 3$$

$$= 7$$

2 Algebra and graphs (continued)

C2.10 Graphs of functions

Notes and examples

- 1 Construct tables of values, and draw, recognise and interpret graphs for functions of the following forms:

- $ax + b$ Linear: 
- $\pm x^2 + ax + b$ Quadratic: 
- $\frac{a}{x}$ ($x \neq 0$) Reciprocal: 

where a and b are integer constants.

① Reciprocal graphs = $\frac{a}{x}$

$a > 0$: 

$a < 0$: 

Eg. $y = 2x + 3$

② For $\frac{a}{x^2}$

$a > 0$: 

$a < 0$: 

- 2 Solve associated equations graphically, including finding and interpreting roots by graphical methods.

e.g. find the intersection of a line and a curve.

To find this, write them both as equal to each other: $2x + 3 = x^2 + 3x + 2$, E.g. $y = x^2 + 3x + 2$

C2.11 Sketching curves

Notes and examples

Recognise, sketch and interpret graphs of the following functions:

- linear
- quadratic.

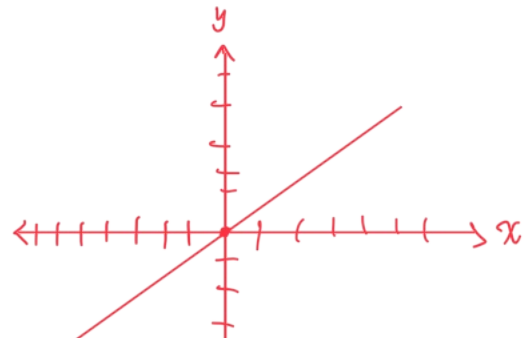
Then solve to find the coordinates of the point/s of intersection.

Knowledge of symmetry and roots is required.

Knowledge of turning points is **not** required.

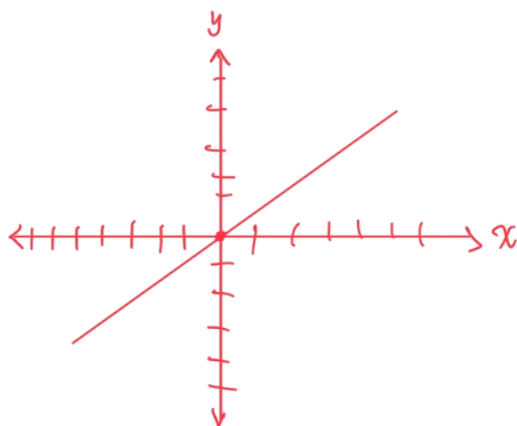
C2.12 Extended content only.

C2.13 Extended content only.



LINEAR

$$y = mx + c$$



QUADRATIC

$$y = ax^2 + bx + c$$

If a is positive: $\cup$ (Smiley face graph)

If a is negative: $\cap$ (Sad face graph)